Analysis of Students' Alcohol Consumption

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Abstract

Portugal, with the highest daily alcohol consumption in Europe (21.4%), presents a significant context for studying adolescent alcohol abuse. This rate is notably higher than neighboring Spain (13.5%) and Italy (12.4%) [1].
This study explores whether Portuguese students' weekday alcohol consumption can be predicted based on their academic performance, social background, and personal habits.
To address this, properly trained Machine Learning models were developed and their predictive effectiveness was evaluated.
The dataset, sourced from Kaggle, originates from a survey conducted among secondary school students enrolled in mathematics and Portuguese courses. It includes a wide range of demographic, academic, and social attributes, such as family background, study habits, extracurricular activities, and health status.
In this study, *Dalc* (weekday alcohol consumption) was chosen as the target variable over *Walc* (weekend alcohol consumption).
The rationale behind this choice is that alcohol consumption during the week could indicate potential academic and behavioral issues, making it a more concerning phenomenon. Conversely, weekend alcohol consumption is likely influenced by social activities and recreational outings, which may not necessarily reflect a problematic drinking pattern.
Data were processed using *KNIME*, applying feature selection, class balancing techniques, and multiple classification models. Additionally, the *R Snippet Node* was used, as there is no a built-in method to evaluate the correlation between categorical variables. To address this, an *R* script was written to compute the normalized χ^2 statistic. After evaluating different approaches, the best-performing model, Random Forest, achieved an accuracy of 95.019 % in the final deployment phase.

1 Introduction

Alcohol abuse among high school students is a significant social issue, with far-reaching implications that extend beyond personal well-being, affecting academic performance and social life. Research in educational settings has shown that adolescent alcohol consumption is often influenced by a combination of social and environmental factors. Notably, studies have shown that students who abuse alcohol are more likely to drink until intoxicated, use alcohol to cope with depression, or to facilitate social interactions [2].
Furthermore, excessive alcohol consumption during adolescence is associated with negative outcomes, such as declining academic performance and an increased likelihood of engaging in risky behaviors [3].
Given the serious repercussions of alcohol abuse in educational environments, there is an urgent need for in-depth investigations into the factors that contribute to this phenomenon.
The objective of this study is to predict alcohol abuse on weekdays by considering various aspects of a student's life, thereby examining their involvement in deviant behaviors.
For the development of the study, KNIME was utilized to implement the modeling process, apply feature selection methods, balance the dataset, and evaluate multiple classification algorithms.

2 Dataset Description

The *Student Alcohol Consumption Dataset* from Kaggle is based on a survey of secondary school students enrolled in Mathematics and Portuguese courses. It consists of two separate datasets, one for each course, but both share the same attributes. The Mathematics dataset contains 395 instances, while the Portuguese dataset includes 649. With a total of 33 attributes, the dataset covers a range of factors, including student demographics, academic performance, family background, social life, behavior, and alcohol consumption.
The key categories and their variables are:

- **Demographic**: sex, age, address, famsize, Pstatus, guardian
- **Academic**: studytime, failures, schoolsup, famsup, paid, G1, G2, G3
- **Family Background**: Medu, Fedu, Mjob, Fjob, reason, famrel
- **Social**: freetime, goout, romantic, activities
- **Behavioral & Health**: health, absences
- **Alcohol Consumption**: Dalc, Walc

More informations regarding the variables' meaning can be found in the Variables' Description appendix.

3 Preprocessing

3.1 Variable Creation

To facilitate distinguishing between students from the two courses before merging the datasets, a new column, *Subject*, was added.
This column assigns "M" for Mathematics students and "P" for Portuguese students, ensuring that each student's course can be clearly identified once the datasets are merged. This distinction is important, as final grades (*G3*) may be influenced by the subject being studied.

3.2 Dataset Creation

The two datasets for Mathematics and Portuguese students were concatenated into a single dataset without altering the column structure, as the columns in both datasets were identical.
Students enrolled in both courses were excluded from the analysis to prevent potential bias. This was necessary because, in such cases, the *G3* variable could be interpreted as the average of final grades from both subjects. Since grades could differ between the subjects, a higher score in one could skew the overall average. To ensure a more accurate and unbiased analysis, we treated the two subjects as independent.

3.3 Variables' Trasformation

The analysis performed using the *Statistics Node* in the workflow revealed that some numerical variables in the dataset should actually be treated as categorical. As a result, the following variables were converted from numerical to categorical:

- Medu
- Fedu
- traveltime
- studytime
- famrel
- freetime
- goout
- Dalc
- Walc
- health

3.4 Missing Values

It was not necessary to apply missing value handling techniques since there are no missing values in the dataset.

3.5 Collinearity

Collinearity was assessed by analyzing the *Correlation Matrix* of the numerical variables, revealing that *G1*, *G2*, and *G3*, which represent the grades of the first semester, the grades of the second semester, and the final grade respectively, are perfectly correlated, as expected. In fact, the variable G3 is a linear combination of G1 and G2.

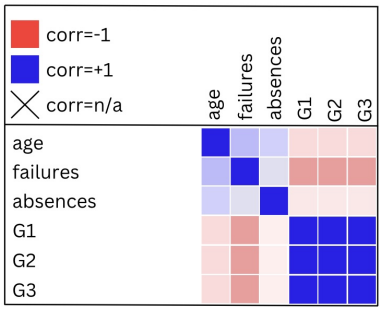

Figure 1: Correlation matrix of numerical variables

Then, all categorical variables were examined. Since KNIME does not provide a method to evaluate the correlation between categorical variables, a Normalized χ^2 Metric R Code was written to compute the normalized χ^2 metric. We chose a threshold of 0.8 for this metric, which was met by all variables in our dataset as it can be seen in the Normalized χ^2 Values. For this reason, all categorical variables were retained.

3.6 Feature Selection

The *Column Filter Node* in KNIME is used to select or exclude columns from a dataset, allowing the removal of unnecessary columns or those that might negatively impact the model's results.
This node was used to remove variables related to semester grades (*G1* and *G2*) and weekend alcohol consumption (*Walc*) to prevent them from biasing the models.

3.7 Outliers

To visualize outliers, a *Box and Whiskers Plot* was created.

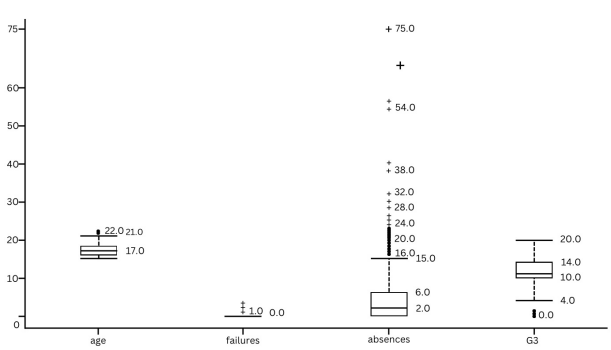

Figure 2: Boxplot, showing the distribution, median, and potential outliers of numerical variables

Observing the distribution of numerical variables, *Absences* exhibited a high number of outliers. To address this issue and ensure that the data is on a comparable scale, the *Normalizer Node* was applied. This node normalizes all numerical variables by scaling them to a standard range, helping prevent variables with large numerical ranges from dominating the model due to their extreme values.

4 Imbalanced Data

To understand the statistical properties of the dataset, the *Statistics Node* was applied, which automatically classified the variables into numerical and categorical groups, allowing for a better understanding of the dataset's structure. Since *Dalc* is the key variable we aim to predict, it is crucial to examine its distribution and assess the balance among different levels of alcohol consumption during weekdays. To visualize this distribution, a *histogram* was used, as it effectively represents the frequency of observations across different categories of *Dalc*. Histograms are particularly useful for identifying patterns, such as skewness or imbalances in the data, helping to determine whether certain consumption levels are over or underrepresented. This insight is essential for guiding the subsequent modeling process.

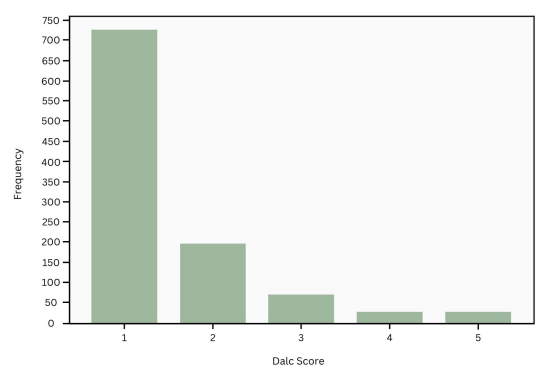

Figure 3: Distribution of students' alcohol consumption levels during weekdays

The histogram highlights a strong imbalance in the distribution of *Dalc*, with the majority of students reporting low alcohol consumption and only a small fraction exhibiting frequent drinking behaviors. This skewed distribution poses a challenge for classification, as the model may be biased toward predicting the majority class, potentially

overlooking patterns associated with higher alcohol consumption levels.
Given the significant impact of excessive alcohol consumption on academic performance, mental health, and overall well-being, it is essential to account for this imbalance in the modeling process.
Addressing this issue through resampling techniques and balancing strategies will be key to ensuring that the model accurately identifies the factors influencing higher alcohol consumption levels and provides meaningful insights for targeted interventions.

4.1 Stratified Sampling

It was decided to use 70% of the data for the training dataset and 30% for the test dataset. Since the dataset exhibits significant class imbalance for the *Dalc* variable, *Stratified Sampling* was applied. This approach was preferred over *Random Sampling*, as the latter could have led to the complete exclusion of less frequent classes. By using *Stratified Sampling*, we ensure that all categories are represented, as this technique best replicates the proportions present in the original dataset.

4.2 Over-sampling

To balance the training dataset, the *SMOTE[4] (Synthetic Minority Over-sampling Technique) Node* was used. It applies an over-sampling approach, where the minority class is over-sampled by generating synthetic examples instead of duplicating existing ones.
Synthetic samples are created as follows:

1. Compute the difference between a given feature vector (sample) and its nearest neighbor.

2. Multiply this difference by a random number between 0 and 1.

3. Add the resulting value to the original feature vector, thereby generating a new synthetic sample.

This method effectively selects a random point along the line segment between two specific features, expanding the decision region of the minority class and making it more general.
SMOTE can thus increase the variety of examples in the minority class without simply duplicating the same data. This brings several significant benefits, including reduction of overfitting and increased generalization.

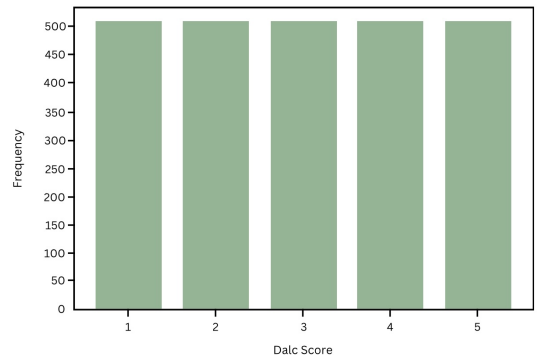

Figure 4: Balanced Training Dataset

5 Classification Models

Various Machine Learning classification techniques were applied to predict weekday alcohol consumption (*Dalc*) among students. These models were grouped into different categories based on their underlying approach.

To train the models, the *Learner Node* was utilized, it handles the model training phase and is specific to each type of machine learning algorithm. It is designed to build the model based on the provided training data.
Predictions were then made using the *Predictor Node*, which enables a pre-trained machine learning model to be used on the test dataset to generate predictions.
The model's performance was evaluated with the *Scorer Node*, whuch is used to evaluate the performance of a machine learning model after it has been trained and applied to a dataset. It was used to assess the predictive accuracy of the model.

5.1 Heuristic Models

These models utilize rule-based decision-making and partition the dataset through tree structures.

- **Decision Trees**: employed to develop interpretable models that predict alcohol consumption based on student attributes. These models use a hierarchical structure, where

internal nodes represent decisions based on specific features, and leaf nodes predict class labels.

  - **J48 algorithm**: an implementation of the C4.5 decision tree, was applied due to its capability to handle both categorical and continuous attributes.
  - **Random Forest**: an ensemble learning method, builds multiple decision trees and combines their predictions for improved accuracy and robustness, making it less prone to overfitting than individual decision trees.

- **Ensemble Trees**: combine multiple models to improve classification performance, aiming to reduce overfitting, increase stability, and enhance generalization.
  - **Gradient Boosting**: sequentially builds models to correct errors made by previous models, offering high accuracy but requiring careful tuning to avoid overfitting.
  - **Bagging**: trains multiple models on different data subsets and combines their predictions to reduce variance and overfitting.

5.2 Probabilistic Models

- **Naive Bayes**: A probabilistic classifier based on Bayes' theorem, assuming feature independence. Both *Naive Bayes* and *Naive Bayes Learner Nodes* were employed in training.
- **Bayesian Network**: were utilized to capture the probabilistic relationships between the variables.
  - The *BayesNet Node* in KNIME was employed, and two learners were applied.
    TANB Learner (Tree-Augmented Naive Bayes) builds a Bayesian network with a tree structure.
    BNC Learner (Bayesian Network Classifier) is a more flexible approach that learns complex relationships between features without a tree structure, offering improved accuracy at the cost of higher computational complexity.
  - **NBTree**: a hybrid model that combines decision trees and Naive Bayes classification.

5.3 Separation Models

Aim to partition or separate the feature space into different regions or classes by finding an optimal decision boundary.

- **Artificial Neural Networks**: computational models consisting of layers of interconnected nodes. Each node processes inputs, applies weights, and passes the output through an activation function to the next layer.
  - **Multilayer Perceptron (MLP)**: consists of multiple layers of neurons, including at least one hidden layer, where each neuron is connected to others via weighted connections. The model's ability to learn from multiple layers of features and refine its predictions makes it a powerful tool for improving classification accuracy and identifying the factors that influence alcohol consumption patterns.

6 Evaluation

To assess the performance of the classification models, a comparative analysis was conducted across different model families based on their Accuracy scores, as well as their respective *Confusion Matrices*.
Accuracy was chosen as evaluation metric due to its straightforwardness and clarity in measuring the overall effectiveness of the models. It provided a direct measure of how well the models classified students into the distinct levels of alcohol consumption.
However, Accuracy was not used in isolation as the sole indicator of model performance. We also examined the confusion matrices to evaluate how well each model handled minority classes and whether misclassifications were systematically affecting certain categories. Generally, models with the best performance exhibited little to no misclassification

of these less frequent classes, reinforcing their reliability beyond just accuracy scores.

6.1 Accuracy Comparison

Heuristic Models
The accuracy of different decision tree models was compared by merging the results of the Decision Tree, J48, and Random Forest models into a final table using the *Compare Accuracy Values Node*.

	Heuristic Models	Accuracy
Decision Trees	Decision tree	0.67
	J48	0.58
	Random forest	0.83
Ensemble Trees	Gradient Boosted	0.77
	Bagging	0.63

Figure 5: Accuracy Comparison in Heuristic Model

The Random Forest achieved the highest accuracy (0.83), making it the best-performing model within this category. Similarly, among the Ensemble Trees, the Gradient Boosted Trees obtained the highest accuracy (0.77). Once the best models from each category were identified, their respective accuracy values were extracted using the *Column Filter Node* and subsequently merged using the *Joiner Node* for further comparison.

Separation Models
The accuracy of the Multilayer Perceptron (MLP), the only model within the Separation Models category, was extracted and joined with the previously merged accuracy values from the Decision Trees and Ensemble Trees.

Separation Models	Accuracy
Multilayer Perceptron	0.78

Figure 6: Accuracy in Separation Model

Probabilistic Models
The accuracy values of the Naive Bayes models and Bayesian Network models were compared.

	Probabilistic Models	Accuracy
Naive Bayes	Naive Bayes	0.50
	Naive Bayes Learner	0.54
Bayesian Network	TANB Learner	0.72
	NB Tree	0.73
	BNC Learner	0.65

Figure 7: Accuracy Comparison in Probabilistic Models

Among the Naive Bayes models, the Naive Bayes Learner achieved the highest accuracy (0.54), while in the Bayesian Network category, the NBTree model performed the best with an accuracy of 0.73. After identifying the best models from both the Naive Bayes and Bayesian Network categories, their respective accuracy values were merged using the *Joiner Node*.

6.2 Best Model

After identifying the best models within each family, a *Joiner (Join all) Node* was performed to create a final table containing the accuracy values for each of the top-performing models from each category.
The accuracy of these models was visualized using a box plot.

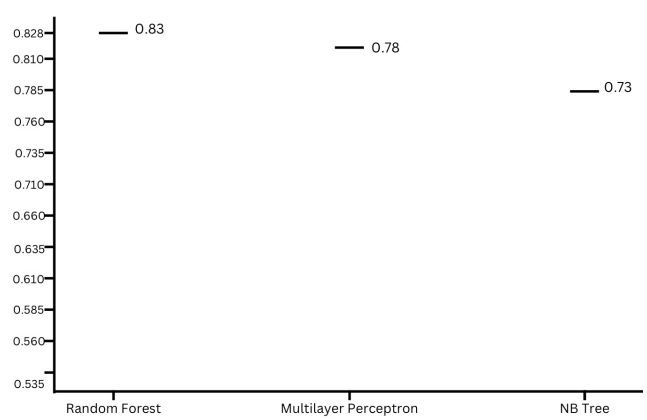

Figure 8: Comparison of the accuracy scores of the top-performing models

The analysis showed that the Random Forest achieved the highest accuracy, which led to its selection as the final model for deployment.

7 Deployment

The *Random Forest* model was extracted using the *Model Writer Node* and reintroduced into the Deployment phase. This node allows the model to be stored in a reusable and shareable state, eliminating the need for retraining each time it is used. One crucial aspect of the model's configuration is the number of trees (*numTrees*), which was set to 50 in this implementation.

Figure 9: Number of trees parameter in the Random Forest model

This parameter plays a fundamental role in determining the model's performance, as it directly influences both accuracy and computational cost. Increasing the number of trees typically enhances stability and reduces variance; however, it also leads to higher computational demands. To ensure a balance between robustness and efficiency, 50 trees were chosen as an optimal trade-off.
With the model properly configured and stored, it was then applied using the *Predictor Node* to generate predictions. These predictions were subsequently evaluated in the *Scorer Node*.

Actual \ Predicted	1	2	5	3	4
1	720	7	0	0	0
2	24	171	1	0	0
5	2	1	23	0	0
3	10	3	0	56	0
4	3	1	0	0	22

Accuracy Overall	Overall Error	Cohen's Kappa	Correctly Classified	Wrongly Classified
95.019%	4.981%	0.891	992	52

Figure 10: Scorer Node's Output

The final dataset evaluation shows an overall accuracy of 95.019%, confirming the robustness of the Random Forest model. The confusion matrix highlights that most predictions are correct, with only 52 misclassified instances out of 1,044. A notable point of attention is the misclassification of class 2 into class 5. However, class 5 (which is less frequent) was misclassified as class 2 (which is more frequent) only once. This suggests that the model does not simply favor the more represented classes, as one would expect in cases of overfitting. Instead, it demonstrates good generalization even for less frequent classes. Additionally, Cohen's Kappa score of 0.891 reinforces the high reliability of the model's predictions. Overall, these results indicate that the model generalizes well and is suitable for deployment.

References

[1] The Portugal News. *Portugal top in Europe for alcohol consumption*. Accessed: 2025-02-01. 2024. URL: https://www.theportugalnews.com/news/2024-04-15/portugal-top-in-europe-for-alcohol-consumption/87944.

[2] Kenneth H. Beck and Terry G. Summons. "Social Context of Alcohol Consumption and Sources of Information Among High School Alcohol Abusers". In: 1986. URL: https://api.semanticscholar.org/CorpusID:140310826.

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[4] Nitesh Chawla et al. "SMOTE: Synthetic Minority Over-sampling Technique". In: *J. Artif. Intell. Res. (JAIR)* 16 (June 2002), pp. 321–357. DOI: 10.1613/jair.953.

Appendix

Variables' Description

Here it's reported a categorized list of the dataset's variables along with a brief description:

- **school**: Student's school (binary: "GP" - Gabriel Pereira, "MS" - Mousinho da Silveira).
- **sex**: Student's gender (binary: "F" - female, "M" - male).
- **age**: Student's age (numeric: from 15 to 22).
- **address**: Student's home address type (binary: "U" - urban, "R" - rural).
- **famsize**: Family size (binary: "LE3" - less or equal to 3, "GT3" - greater than 3).
- **Pstatus**: Parent's cohabitation status (binary: "T" - living together, "A" - apart).
- **Medu**: Mother's education level (numeric: 0 - none, 1 - primary, 2 - 5th to 9th grade, 3 - secondary, 4 - higher education).
- **Fedu**: Father's education level (numeric: same scale as Medu).
- **Mjob**: Mother's job (nominal: "teacher", "health", "services" , "at_home", "other").
- **Fjob**: Father's job (nominal: same categories as Mjob).
- **reason**: Reason for choosing the school (nominal: "home" - close to home, "reputation" - school reputation, "course" - course preference, "other").
- **guardian**: Student's legal guardian (nominal: "mother", "father", "other").
- **traveltime**: Home-to-school travel time (numeric: 1 - $<$ 15 minutes, 2 - 15 to 30 minutes, 3 - 30 minutes to 1 hour, 4 - $>$ 1 hour).
- **studytime**: Weekly study time (numeric: 1 - $<$ 2 hours, 2 - 2 to 5 hours, 3 - 5 to 10 hours, 4 - $>$ 10 hours).
- **failures**: Number of past class failures (numeric: from 0 to 4).
- **schoolsup**: Extra educational support provided by the school (binary: "yes" - yes, "no" - no).

- **famsup**: Extra educational support provided by the family (binary: “yes” - yes, “no” - no).
- **paid**: Attended extra paid classes in Math or Portuguese (binary: “yes” - yes, “no” - no).
- **G1**: First period grade (numeric: from 0 to 20).
- **G2**: Second period grade (numeric: from 0 to 20).
- **G3**: Final grade (numeric: from 0 to 20, target variable for academic prediction).
- **freetime**: Free time after school (numeric: 1 - very low, 5 - very high).
- **goout**: Frequency of going out with friends (numeric: 1 - very low, 5 - very high).
- **romantic**: In a romantic relationship (binary: “yes” - yes, “no” - no).
- **activities**: Participation in extracurricular activities (binary: “yes” - yes, “no” - no).
- **nursery**: Attended nursery school (binary: “yes” - yes, “no” - no).
- **higher**: Plans to pursue higher education (binary: “yes” - yes, “no” - no).
- **internet**: Internet access at home (binary: “yes” - yes, “no” - no).
- **famrel**: Quality of family relationships (numeric: 1 - very bad, 5 - excellent).
- **health**: Current health status (numeric: 1 - very bad, 5 - very good).
- **absences**: Number of school absences (numeric: from 0 to 93).
- **Dalc**: Weekday alcohol consumption (numeric: 1 - very low, 5 - very high).
- **Walc**: Weekend alcohol consumption (numeric: 1 - very low, 5 - very high).

Normalized χ^2 Metric R Code

```
knime.out <- knime.in
library(dplyr)
library(plyr)
combos <- combn(ncol(knime.out), 2)
adply(combos, 2, function(x) {
  test <- chisq.test(knime.out[, x[1]], knime.out[, x[2]])

  out <- data.frame("Row" = colnames(knime.out)[x[1]],
                    "Column" = colnames(knime.out[x[2]]),
                    "Chi.Square" = round(test$statistic, 3),
                    "df" = test$parameter,
                    "p.value" = round(test$p.value, 3)
  )
  return(out)
})

# Chi-squared normalizzato
combos <- combn(ncol(knime.out), 2)
adply(combos, 2, function(x) {
  test <- chisq.test(knime.out[, x[1]], knime.out[, x[2]])
  tab <- table(knime.out[, x[1]], knime.out[, x[2]])
  out <- data.frame("Row" = colnames(knime.out)[x[1]],
                    "Column" = colnames(knime.out[x[2]]),
                    "Chi.Square" = round(test$statistic, 3),
                    "df" = test$parameter,
                    "p.value" = round(test$p.value, 3),
                    "n" = sum(tab),
                    "u1" = length(unique(knime.out[, x[1]])) - 1,
                    "u2" = length(unique(knime.out[, x[2]])) - 1,
                    "nMinu1u2" = n * min(u1, u2),
                    "Chi.Square norm" = test$statistic / (n * min(u1, u2))
  )
  return(out)
})
```

Normalized χ^2 Values

Row	Column	Normalized χ^2
activities	freetime	1.23E-01
traveltime	Subject	1.15E-01
paid	freetime	1.07E-01
studytime	famrel	9.47E-02
famsup	famrel	8.37E-02
paid	famrel	8.31E-02
studytime	paid	7.60E-02
studytime	internet	6.75E-02
schoolsup	goout	6.44E-02
schoolsup	paid	4.85E-02
studytime	nursery	4.69E-02
higher	romantic	4.16E-02
activities	higher	4.11E-02
paid	internet	4.03E-02
studytime	activities	3.75E-02
paid	Subject	3.74E-02
nursery	romantic	3.69E-02
famsup	activities	3.40E-02
studytime	higher	2.85E-02
activities	romantic	2.81E-02

Figure 11: Twenty Highest χ^2 Values